

Normalization & Indexing - Quick Revision Cheat Sheet

QUICK REVISION & EXAM CHEAT SHEET

Subject: Database Management Systems | Chapter: Chapter 2: Normalization, Relational Algebra & Indexing | Academic Year: 2026-2027

Normal Forms Decision Matrix

- * 1NF: Atomic values only. No repeating groups.
- * 2NF: 1NF + No partial dependency (non-prime cannot depend on proper subset of candidate key).
- * 3NF: 2NF + No transitive dependency. For $X \rightarrow Y$: X is superkey OR Y is prime attribute.
- * BCNF: For every dependency $X \rightarrow Y$, X MUST be a superkey (no exceptions for prime attributes).

Relational Algebra Operators

- * Selection (σ): Filters tuples (rows). Example: $\sigma_{\text{salary} > 50000}(\text{Employee})$.
- * Projection (π): Extracts specific attributes (columns). Example: $\pi_{\{\text{name}, \text{role}\}}(\text{Employee})$.
- * Natural Join (bowtie): Equijoin on all common attribute names followed by projection.

B-Tree versus B+ Tree Indexing

- * B-Tree stores data pointers in both internal and leaf nodes.
- * B+ Tree stores data pointers ONLY in leaf nodes; leaves are linked for $O(\log n)$ + range scan performance.
- * B+ Tree internal nodes hold more keys per disk block, resulting in lower tree height and fewer disk I/O operations.